
Software Cost Estimation Practice (Function Point)

These three practices follow the same four-step model as the sample problem:

1. Find the Function Points (FP)
 2. Find the Estimated LOC
 3. Find the Cost of Software
 4. Find the Effort
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Reference Tables (needed for every solution)

Table A — IFPUG Complexity Weights

Function Type	Low	Average	High
External Input (EI) — data entering the system	3	4	6
External Output (EO) — data leaving the system (reports, screens, printouts)	4	5	7
External Inquiry (EQ) — input/output pair that retrieves data, no update	3	4	6
Internal Logical File (ILF) — data maintained inside the system	7	10	15
External Interface File (EIF) — data used by the system but maintained by another system	5	7	10

Table B — Average LOC per Function Point (by language)

Language	LOC/FP
Assembly	320
C	128
COBOL / FORTRAN	105
Pascal	91
C++ / Java	53
Visual C++	34
Visual Basic	32
4GL	20

Table C — Value Adjustment Factor (VAF)

$$\text{VAF} = 0.65 + (0.01 \times \text{TDI})$$

TDI (Total Degree of Influence) = sum of ratings (0–5) of the 14 General System Characteristics (GSCs): data communications, distributed data processing, performance, heavily used configuration, transaction rate, online data entry, end-user efficiency, online update, complex processing, reusability, ease of installation, ease of operation, portability, facilitate change. TDI ranges 0–70, so VAF ranges 0.65–1.35.

Core Formulas

- Unadjusted Function Points: $\text{UFP} = \sum (\text{count} \times \text{weight})$
- Adjusted Function Points: $\text{FP} = \text{UFP} \times \text{VAF}$
- Estimated LOC: $\text{LOC} = \text{FP} \times (\text{LOC/FP for the language})$
- Cost of Software: $\text{Cost} = \text{LOC} \times (\$/\text{LOC})$
- Effort: $\text{Effort (person-months)} = \text{LOC} \div (\text{LOC/person-month})$

PRACTICE 1

Problem

A small HR system has the following functions:

No. of Functions	Description	Weight
2	Enter a new employee record via a form	Low
1	Store employee records in the system's own database	Average
3	Display employee details on screen	Low
2	Search for an employee by ID	Average

Given: TDI = 40 Language = C
 LOC/Person-month = 300 \$/LOC = 15

Find: (1) Function Points
 (2) Estimated LOC
 (3) Cost of software
 (4) Effort

PRACTICE 2

Problem

A retail management system has the following functions:

No. of Functions	Description	Weight
3	Read configuration files maintained by another application	Low
2	Update a customer profile through a web form	Average
4	Generate a monthly sales report on screen	Average
2	Generate a monthly sales report as a PDF file	High
3	Check order status by order number	Low
1	Maintain product inventory data within the system	High
2	Import a price list from a supplier's system	Average

Given: TDI = 48, Language = Visual C++,
LOC/Person-month = 360, \$/LOC = 24

Find: (1) Function Points,
(2) Estimated LOC,
(3) Cost of software,
(4) Effort.
